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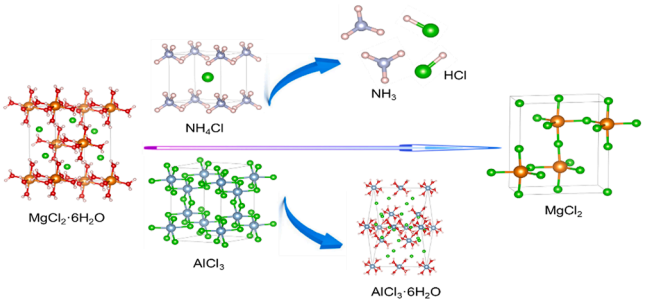
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Significant reduction in the dehydration temperature of magnesium chloride hexahydrate through the addition of $AlCl_3$ and NH_4Cl

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GRAPHICAL ABSTRACT



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ABSTRACT

The electrolytic production of magnesium metal requires anhydrous $MgCl_2$ with a high-purity, which is usually prepared through the dehydration of $MgCl_2{\cdot}6H_2O$ at a relatively high temperature. The Mg^{2+} cations in $MgCl_2{\cdot}6H_2O$ have a high affinity for water and the complete dehydration of $MgCl_2{\cdot}6H_2O$ is difficult due to the hydrolysis of the hydrated Mg^{2+} cations. The hydrolysis usually results in the formation of unwanted oxy-compounds of magnesium. In this study, the effects of the addition of $AlCl_3$ and NH_4Cl on the dehydration of $MgCl_2{\cdot}6H_2O$ were investigated by comparing the thermal decomposition processes of $MgCl_2{\cdot}6H_2O$, $MgCl_2{\cdot}6H_2O$/$AlCl_3$, and $MgCl_2{\cdot}6H_2O$/NH_4Cl, using Raman, FT-IR, TG-DSC, XRD techniques and theoretical calculations. It was found that the addition of $AlCl_3$ or NH_4Cl significantly reduces the dehydration energy barrier of $MgCl_2{\cdot}6H_2O$, decreases the dehydration temperature, and suppresses the hydrolysis of $MgCl_2{\cdot}6H_2O$. Anhydrous $AlCl_3$ can extract water from $MgCl_2{\cdot}6H_2O$ at room temperature. The interactions between Al^{3+} or NH_4^+ and H_2O, together with the substitution of Cl^- for H_2O, result in the easier production of anhydrous $MgCl_2$. During the dehydration of $MgCl_2{\cdot}6H_2O$, several impurities including MgO are generated, indicating that the environment remains to be an important controlling factor for the dehydration of $MgCl_2{\cdot}6H_2O$. The results obtained would be helpful for

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understanding the dehydration processes and mechanisms of $MgCl_2·6H_2O$ under the assistance of $AlCl_3$ and NH_4Cl, and for exploring new techniques for $MgCl_2·6H_2O$ dehydration.

1. Introduction

Anhydrous $MgCl_2$ is widely used in high-temperature resistant materials and in the electrolytic production of magnesium metal [1,2]. Electrical and automotive industries make significant use of magnesium alloys due to their exceptional durability and lightweight properties [3–5]. $MgCl_2·6H_2O$ must be totally dehydrated to produce anhydrous $MgCl_2$, which is then used in the molten salt electrolysis process to produce magnesium metal. The electrolysis process is negatively impacted by the presence of MgO created from the hydrolysis of $MgCl_2$. Thus, the complete dehydration of $MgCl_2·6H_2O$ without $MgCl_2$ hydrolysis has attracted more and more attention [6]. During dehydration, $MgCl_2·6H_2O$ initially melts into its molten salt hydrate [7], and then gradually loses the H_2O molecules to transform into $MgCl_2$ hydrates with different levels of hydration: $MgCl_2·4H_2O$, $MgCl_2·2H_2O$, and $MgCl_2·H_2O$ [1,8,9]. Partial dehydration from $MgCl_2·6H_2O$ or $MgCl_2·4H_2O$ to $MgCl_2·2H_2O$ is relatively easy. By contrast, further dehydration from $MgCl_2·2H_2O$ to $MgCl_2·H_2O$ and the complete dehydration from $MgCl_2·2H_2O$ or $MgCl_2·H_2O$ to anhydrous $MgCl_2$ is difficult and the process is primarily conducted under HCl protective atmosphere to prevent $MgCl_2·2H_2O$ and $MgCl_2·H_2O$ from hydrolysis [1]. Consequently, the HCl atmosphere protection method is extensively used in the production of anhydrous $MgCl_2$. However, there are shortcomings related to this approach. The dehydration temperature is relatively high and corrosive tail gases consisting of HCl, H_2O vapor are emitted. Therefore, lowering the dehydration temperature of $MgCl_2·6\ H_2O$ and preventing it from hydrolysis at the same time have drawn more and more attention.

The double salt method is an acceptable alternative approach for preparing anhydrous $MgCl_2$. Magnesium chloride can form potassium and ammonium carnallite double salts with potassium and ammonium chloride, respectively. Potassium and ammonium carnallites can be easily dehydrated at a relatively low temperature and are less prone to hydrolysis. The chemical environment of magnesium cations in both potassium and ammonium carnallites are more helpful for the complete dehydration of the hydrates and for restriction on the hydrolysis [10,11].

Although the formation of double salts is beneficial for dehydration, the preparation of double salts requires some special consideration [12,13]. In this study, an evaluation was conducted to examine the effects of $AlCl_3$ and NH_4Cl addition on the removal of H_2O from $MgCl_2·6H_2O$. Due to the higher hydration energy of Al^{3+} compared to Mg^{2+}, it is speculated that anhydrous $AlCl_3$ may capture H_2O molecules from hydrated Mg^{2+} and promote the dehydration of $MgCl_2·6H_2O$ during heating [14,15]. This research primarily examines the impact of $AlCl_3$ and NH_4Cl on the dehydration process of $MgCl_2·6H_2O$, while excluding the influence of the atmosphere. DFT theoretical calculations are also used to study the mechanism for the dehydration of $MgCl_2·6H_2O$ under the influence of $AlCl_3$ and NH_4Cl.

2. Experimental section and calculations

2.1. Materials

$MgCl_2·6H_2O$ (AR, $\geq$ 99.99 %) was purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Anhydrous $AlCl_3$ (AR, $\geq$ 99.99 %) and NH_4Cl (AR, $\geq$ 99.99 %) were purchased from Shanghai Macklin Biochemical Technology Co., Ltd. All reagents were used without further purification.

2.2. General procedures for the preparation of the mixed salts

The mixed salt of $MgCl_2·6H_2O$ and $AlCl_3$ was prepared by mixing $MgCl_2·6H_2O$ and $AlCl_3$ powders in a molar ratio of $MgCl_2·6H_2O$:$AlCl_3$ being 3:1. The mixture was grinded in an agate crucible for 10 min at room temperature. The mixed salt of $MgCl_2·6H_2O$ and NH_4Cl was prepared following a similar procedure. The mixed salts of $MgCl_2·6H_2O$ with $AlCl_3$ and NH_4Cl were denoted as $MgCl_2·6H_2O/AlCl_3$ and $MgCl_2·6H_2O/NH_4Cl$, respectively.

2.3. Characterization

The Raman spectra were performed on a confocal Raman Microscope (DXR, Thermo Fisher Scientific Inc.) with an excitation laser wavelength of 532 nm and an incident power of 10 mW. The TG-DSC analyses were performed under a N_2 atmosphere using a simultaneous thermal analyzer (NETZSCH, STA449F3) at a heating rate of 10 °C/min. The in-situ high-temperature XRD experiments were performed using an X-ray diffractometer (BRUKER, D8 Discover) under vacuum. The samples of $MgCl_2·6H_2O$, $MgCl_2·6H_2O/AlCl_3$, and $MgCl_2·6H_2O/NH_4Cl$ were subjected to heating at a constant rate of 5 °C/min, and XRD data of all samples was acquired every 10 °C from 25°C to 400°C. The FT-IR spectra were obtained using a VERTEX 70 spectrometer (KBr method, Bruker, Germany) with a resolution of 2 cm^{-1}.

2.4. Calculations

To identify the potential for energy changes and reaction directions in $MgCl_2·6H_2O/AlCl_3$ and $MgCl_2·6H_2O/NH_4Cl$ systems, the changes in internal energy for the conversion reactions between $MgCl_2$ and MgO in these two systems were calculated using Density Functional Theory (DFT). The calculations were further optimized with the lowest energy at the B3LYP/6-31g(d,p) level of caltheory for both $MgCl_2·6H_2O/AlCl_3$ and $MgCl_2·6H_2O/NH_4Cl$ systems. The Gaussian-16 (Revision C.01)/ Vienna Ab-initio Simulation Package (VASP) software was used for all these DFT computations for the molecular and periodic crystal models, respectively. The gradient-corrected Perdew-Burke-Ernzerhof (GGA-PBE) function was adopted to describe electron exchange and correlation energy. The cutoff energy for plane-wave basis set was set as 600 eV, and the total energy convergence was set to be lower than 1×10^{-5} eV, with the force convergence set at 0.02 eV/Å for geometry optimization. The Gaussian smearing of 0.2 eV was used for geometry optimization.

3. Results and discussion

3.1. Raman and FT-IR analyses

$MgCl_2·6H_2O$ contains $[Mg(H_2O)_6]^{2+}$ octahedron units that are well-organized as a result of hydrogen bonding between H_2O and Cl^-, as shown in Fig. 1(a). The effect of $AlCl_3$ and NH_4Cl on the coordinated H_2O in $MgCl_2·6H_2O$ was investigated. Fig. 1(a) depicts the Raman spectra of $MgCl_2·6H_2O$, $AlCl_3·6H_2O$, NH_4Cl, $MgCl_2·6H_2O/AlCl_3$, and $MgCl_2·6H_2O/NH_4Cl$. Each spectrum covers the entire wavenumber range (100–3800 cm^{-1}). The vibration related with water is mainly recognized in the high frequency range (2800–3600 cm^{-1}). The Raman spectrum of the mixture of $MgCl_2·6H_2O$ and $AlCl_3$ after grinding, related with water within the range of 2800–3600 cm^{-1}, shows obvious similarity to that of $AlCl_3·6H_2O$, without any similarity to that of $MgCl_2·6H_2O$. Such a result indicates that anhydrous $AlCl_3$ can extract water from $MgCl_2·6H_2O$ according to the following equation:

$MgCl_2{\cdot}6H_2O + AlCl_3 = AlCl_3{\cdot}6H_2O + MgCl_2$

The Raman peaks of $AlCl_3{\cdot}6H_2O$ appeared in the Raman spectra of $MgCl_2{\cdot}6H_2O/AlCl_3$. $MgCl_2{\cdot}6H_2O/AlCl_3$ exhibits a relatively rich spectrum in the low-frequency region (100–500 cm^{-1}), with the most intense feature corresponding to the υ(Al-Cl) symmetric stretch at 308 cm^{-1} [16]. The υ(A1-O), υ(Al-Cl), δ(A1-O-A1), and δ(A1-OH) modes for the $MgCl_2{\cdot}6H_2O/AlCl_3$ mixture were identified [17,18]. However, the Raman peaks of O-H vibration in $MgCl_2{\cdot}6H_2O/NH_4Cl$ remain mostly unaltered in the Raman spectra. Fig. 1(b) demonstrates that the FT-IR spectra of N-H^+ stretching (3300–3030 cm^{-1}) and bending (1430–1390 cm^{-1}) vibration peaks of NH_4^+ in NH_4Cl decrease in intensity for the mixture of $MgCl_2{\cdot}6H_2O/NH_4Cl$. It indicates that there are interactions between NH_4^+ and H_2O molecules in $MgCl_2{\cdot}6H_2O$ after the mixture of $MgCl_2{\cdot}6H_2O$ and NH_4Cl was grinded. Thus, the coordinated H_2O in $MgCl_2{\cdot}6H_2O$ can interact with both anhydrous $AlCl_3$ and NH_4Cl.

3.2. Thermal analyses

Fig. 2 illustrates the TG/DTG and DSC characteristics of $MgCl_2{\cdot}6H_2O$, $MgCl_2{\cdot}6H_2O/AlCl_3$, and $MgCl_2{\cdot}6H_2O/NH_4Cl$ in the temperature range of 25–300 °C. The thermal decomposition of $MgCl_2{\cdot}6H_2O$, $MgCl_2{\cdot}6H_2O/AlCl_3$, and $MgCl_2{\cdot}6H_2O/NH_4Cl$ is a complicated multistep process, including overlapping melting, dehydration, hydrolysis, etc. The process shows both endothermic effect and mass loss (mainly coordinate water loss and HCl release). The TG analyses show that at the temperature range of 50–250 °C, the percent weight loss of both $MgCl_2{\cdot}6H_2O/AlCl_3$ and $MgCl_2{\cdot}6H_2O/NH_4Cl$ is greater than that of $MgCl_2{\cdot}6H_2O$.

For $MgCl_2{\cdot}6H_2O$, its dehydration below 200 °C with a total wight loss of 36.1 % (theoretical total wight loss 35.4 %) is correspond to the loss of four water molecules to the state of the dihydrate with intermediate formation of the tetrahydrate. Above 200 °C further dehydration occurs, accompanied by hydrolysis and the release of HCl gas. The HCl release is evident at about 275 °C. These results are generally consistent with the previous work [12]. The endothermic peaks at about 130 and 175 °C all correspond to the dehydration. And the endothermic peaks at about 225 and 280 °C mainly correspond to the hydrolysis of $MgCl_2$ and the decomposition of the hydrolysis products, respectively [19].

For $MgCl_2{\cdot}6H_2O/AlCl_3$, the wight loss at the temperature range of 50–200 °C, and the obvious endothermic peak at around 175 °C is related with the dehydration and hydrolysis of $AlCl_3{\cdot}6H_2O$ [15]. Based on the above Raman and FT-IR studies, after mixing and grinding the mixture of $MgCl_2{\cdot}6H_2O$ and $AlCl_3$, almost all the $MgCl_2{\cdot}6H_2O$ changes into $MgCl_2$, with the production of $AlCl_3{\cdot}6H_2O$.

For $MgCl_2{\cdot}6H_2O/NH_4Cl$, the wight loss below 200 °C is mainly due to the dehydration of $MgCl_2{\cdot}6H_2O$, corresponding to the three endothermic peaks at about 85, 140, 175 °C. The endothermic peaks at about 225 and 275 °C are mainly related with the hydrolysis of $MgCl_2$ and the thermal decomposition of NH_4Cl, respectively. The hydrated water is more easily released as compared to $MgCl_2{\cdot}6H_2O$, because NH_4Cl weakens the interactions between Mg^{2+} and water though the hydrogen bonding between NH_4^+ and water [11]. Furthermore, ammonium chloride is decomposed into HCl and NH_3 at higher temperatures, and HCl and NH_3 can prevent the hydrolysis of magnesium chloride [20].

Based on the thermal analyses, below 160 °C, $MgCl_2{\cdot}6H_2O/NH_4Cl$ losses wight faster than $MgCl_2{\cdot}6H_2O/AlCl_3$. In $MgCl_2{\cdot}6H_2O/NH_4Cl$, the interactions between Mg^{2+} and water is weakened because of the hydrogen bonding between NH_4^+ and water. Therefore, crystalized water in $MgCl_2{\cdot}6H_2O/NH_4Cl$ is easier to release. On the contrary, in $MgCl_2{\cdot}6H_2O/AlCl_3$, after the transfer of water from $MgCl_2{\cdot}6H_2O$ to $AlCl_3$ with the formation of $AlCl_3{\cdot}6H_2O$, although some $MgCl_2{\cdot}6H_2O$ is dehydrolyzed, the crystalized water in $AlCl_3{\cdot}6H_2O$ is not so easy to release. When the temperature is above 160 °C, the dehydration and hydrolysis of $AlCl_3{\cdot}6H_2O$ become apparent with an obvious wight loss [15]. Therefore, the TG curves of $MgCl_2{\cdot}6H_2O/NH_4Cl$ and $MgCl_2{\cdot}6H_2O/AlCl_3$ have a crossing point at about 160 °C.

3.3. In-situ high-temperature XRD analyses

Based on the Raman, FT-IR, and TG-DSC analyses, it can be inferred that the thermal degradation of $MgCl_2{\cdot}6H_2O$ can be affected by both $AlCl_3$ and NH_4Cl. The in-situ high-temperature XRD tests were performed to explore the influence of $AlCl_3$ and NH_4Cl on the dehydration process in a vacuum environment, to further characterize the formation process of anhydrous $MgCl_2$ and by-products.

The solid $MgCl_2{\cdot}6H_2O$ hydrate progressively melts into a molten salt hydrate when heated [21]. However, $MgCl_2{\cdot}6H_2O$ still exhibits a strong peak at 120 °C, indicating no significant melting state under vacuum conditions, as shown in Fig. 3(a). With the increase in temperature, the characteristic peak intensity of $MgCl_2{\cdot}6H_2O$ gradually decreases. $MgCl_2{\cdot}6H_2O$ gradually dehydrates to form $MgCl_2{\cdot}2H_2O$ at 80 °C. With further increase in temperature, significant anhydrous $MgCl_2$ and MgO phases appear at 240 °C.

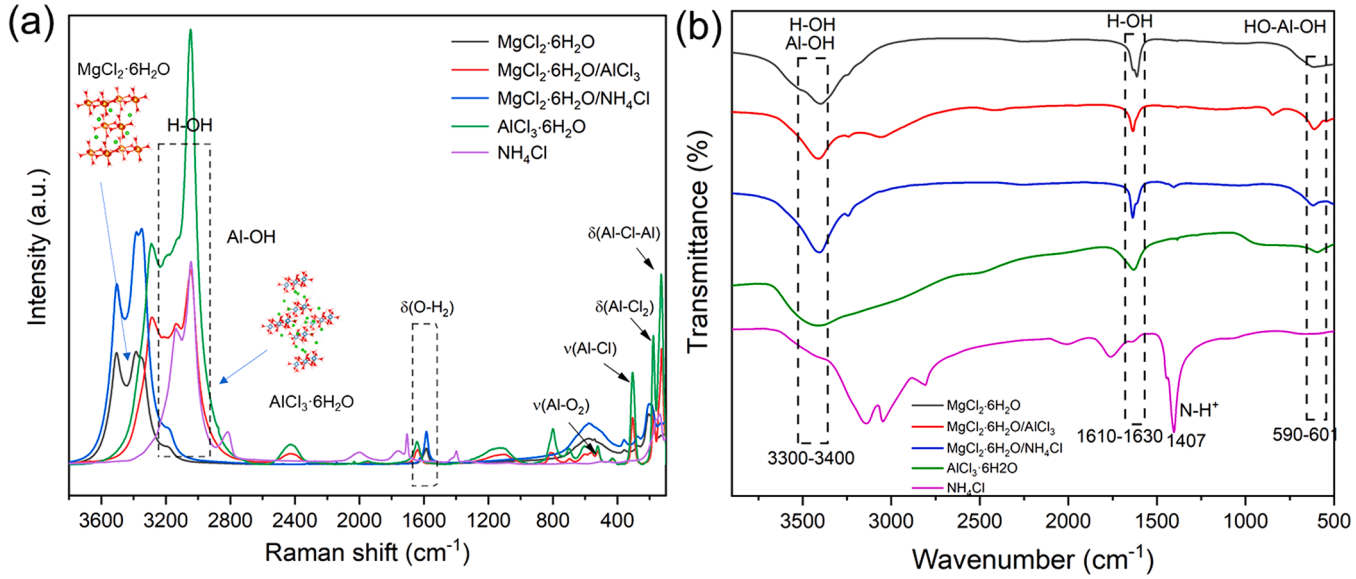


Fig. 1. Raman and FT-IR spectra of $MgCl_2{\cdot}6H_2O$, $MgCl_2{\cdot}6H_2O/AlCl_3$, $MgCl_2{\cdot}6H_2O/NH_4Cl$, $AlCl_3{\cdot}6H_2O$, and NH_4Cl.

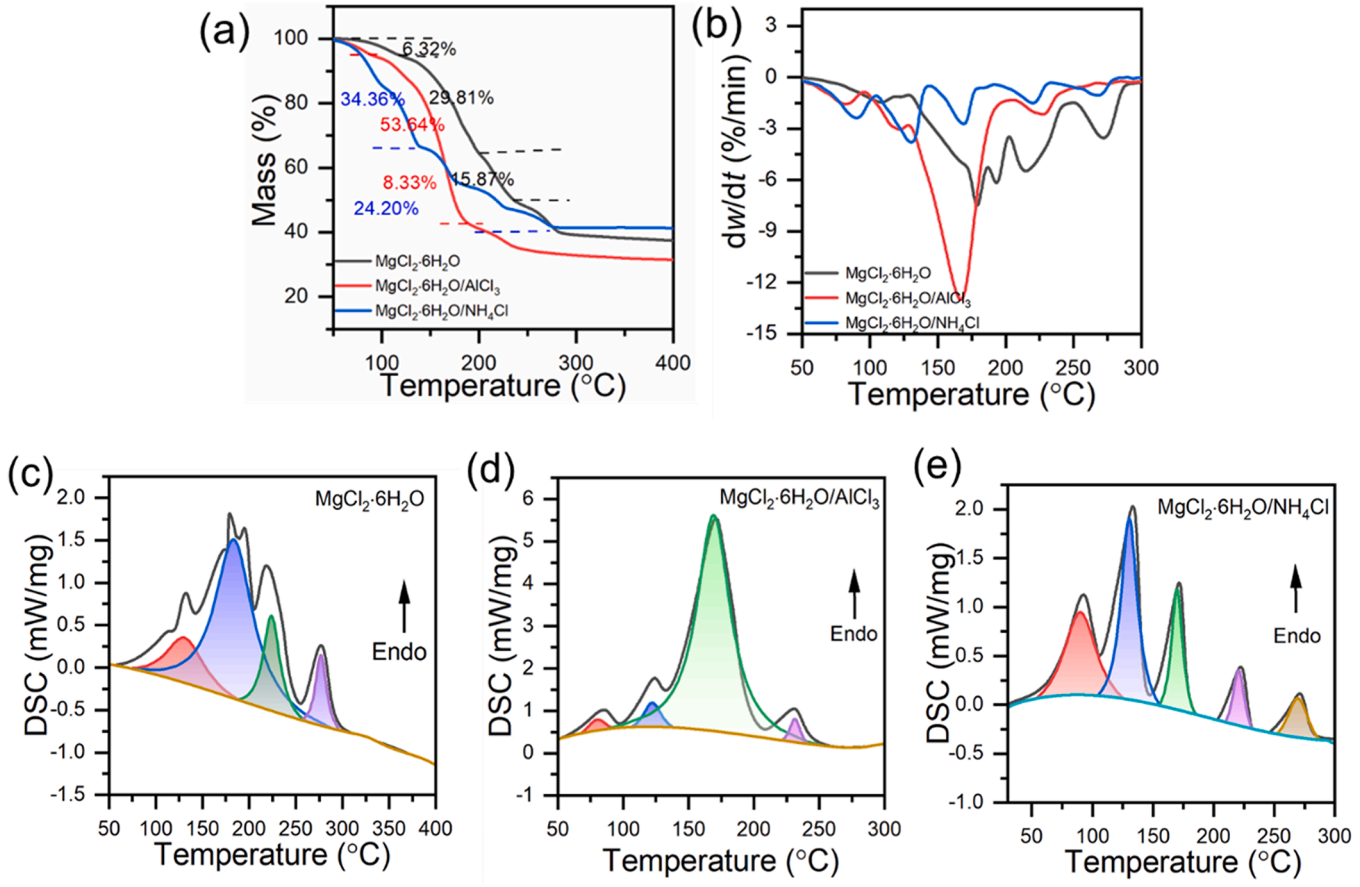


Fig. 2. TG/DTG and DSC analyses of $MgCl_2{\cdot}6H_2O$, $MgCl_2{\cdot}6H_2O/AlCl_3$, and $MgCl_2{\cdot}6H_2O/NH_4Cl$: (a, b) TG and DTG curves; (c, d, e) DSC curves.

Anhydrous $AlCl_3$ has a strong ability to bind H_2O. The XRD contour map of $MgCl_2{\cdot}6H_2O/AlCl_3$ shown in Fig. 3(b) demonstrates that anhydrous $AlCl_3$ can extract the coordinated H_2O from $MgCl_2{\cdot}6H_2O$ to generate anhydrous $MgCl_2$ at 25 °C, confirming the corresponding results of Raman studies. When the temperature is above 120 °C, the distinctive XRD peaks of $MgCl_2{\cdot}6H_2O$ and $AlCl_3$ progressively vanish, and the peak intensity of anhydrous $MgCl_2$ increases. The byproduct peaks of AlOOH and $Al(OH)_3$ appear at 100 °C. Clear Al_2O_3 peaks were observed at 160 °C [15].

For $MgCl_2{\cdot}6H_2O/NH_4Cl$, the peak of anhydrous $MgCl_2$ begins to appear at 70 °C (Fig. 3(c)). NH_3 and HCl produced by the decomposition of NH_4Cl cannot totally protect the $MgCl_2$ hydrates from hydrolyzing under vacuum. Multiple crystal forms of MgO are generated at 120 °C.

Due to the interactions between Al^{3+} or NH_4^+ and H_2O, together with the substitution of Cl^- for H_2O, the addition of $AlCl_3$ or NH_4Cl significantly reduces the generation temperature of anhydrous $MgCl_2$. In general, $AlCl_3$ and NH_4Cl can promote the dehydration of $MgCl_2{\cdot}6H_2O$, generating anhydrous $MgCl_2$. At the same time, various impurities such as $Mg(OH)_2$, MgO, and Al_2O_3 are generated. Hence, the environment remains to be a primary controlling factor for the dehydration of $MgCl_2{\cdot}6H_2O$.

3.4. Calculations

According to the in-situ high-temperature XRD patterns in Fig. 3, after mixing and grinding the mixture of $MgCl_2{\cdot}6H_2O/AlCl_3$, there are $MgCl_2{\cdot}6H_2O$, anhydrous $MgCl_2$, $AlCl_3{\cdot}6H_2O$, and other crystalline hydrates in the system. During heating, there are several reactions related with dehydration and hydrolysis. An important reaction is the conversion reaction between MgO and $MgCl_2$ as shown in Fig. 4(a). Based on our calculations, the internal energy of the products is lower than that of the reactants, with a ΔE of − 260.2 kcal/mol. For reaction (a), the calculation by using VASP with periodic boundary conditions was also performed. The result is qualitatively consistent with the molecular result. The negative energy difference indicates that the component combination in right side with a lower energy is more stable and likely to exist during the reaction process. Therefore, the calculation result shows that the combination with $MgCl_2$ has a lower energy, which means that in the presence of $AlCl_3$ it is feasible the stable of $MgCl_2$ and $MgCl_2$ is less likely to hydrolyze during the heating and dehydration process.

The dehydration of $MgCl_2{\cdot}6H_2O/NH_4Cl$ also involves a similar important reaction as shown in Fig. 4(b). Due to the lower internal energy of the products compared to the reactants (ΔE = -116.0 kcal/mol), the inclusion of NH_4Cl in the dehydration procedure of $MgCl_2{\cdot}6H_2O$ will be helpful for the dehydration of $MgCl_2{\cdot}6H_2O$ and for the suppression of the formation of MgO.

4. Conclusions

The effects of $AlCl_3$ and NH_4Cl on the dehydration of $MgCl_2{\cdot}6H_2O$ were comprehensively studied. In-situ high-temperature XRD tests indicate that $AlCl_3$ and NH_4Cl can effectively reduce the dehydration temperature, and in a vacuum environment anhydrous $MgCl_2$ is generated at 25 and 80 °C, respectively. Such temperatures at which anhydrous $MgCl_2$ is produced are much lower than that of the reported works. The Raman spectrum of the mixture of $MgCl_2{\cdot}6H_2O$ and $AlCl_3$ after grinding, related with water within the range of 2800–3600 cm^{-1}, shows obvious similarity to that of $AlCl_3{\cdot}6H_2O$, without any similarity to that of $MgCl_2{\cdot}6H_2O$. Such a result indicates that anhydrous $AlCl_3$ can extract the coordinated H_2O from $MgCl_2{\cdot}6H_2O$ to generate anhydrous $MgCl_2$. The TG analyses show that at the temperature range of 100–250 °C, the percent weight loss of both $MgCl_2{\cdot}6H_2O/AlCl_3$ and

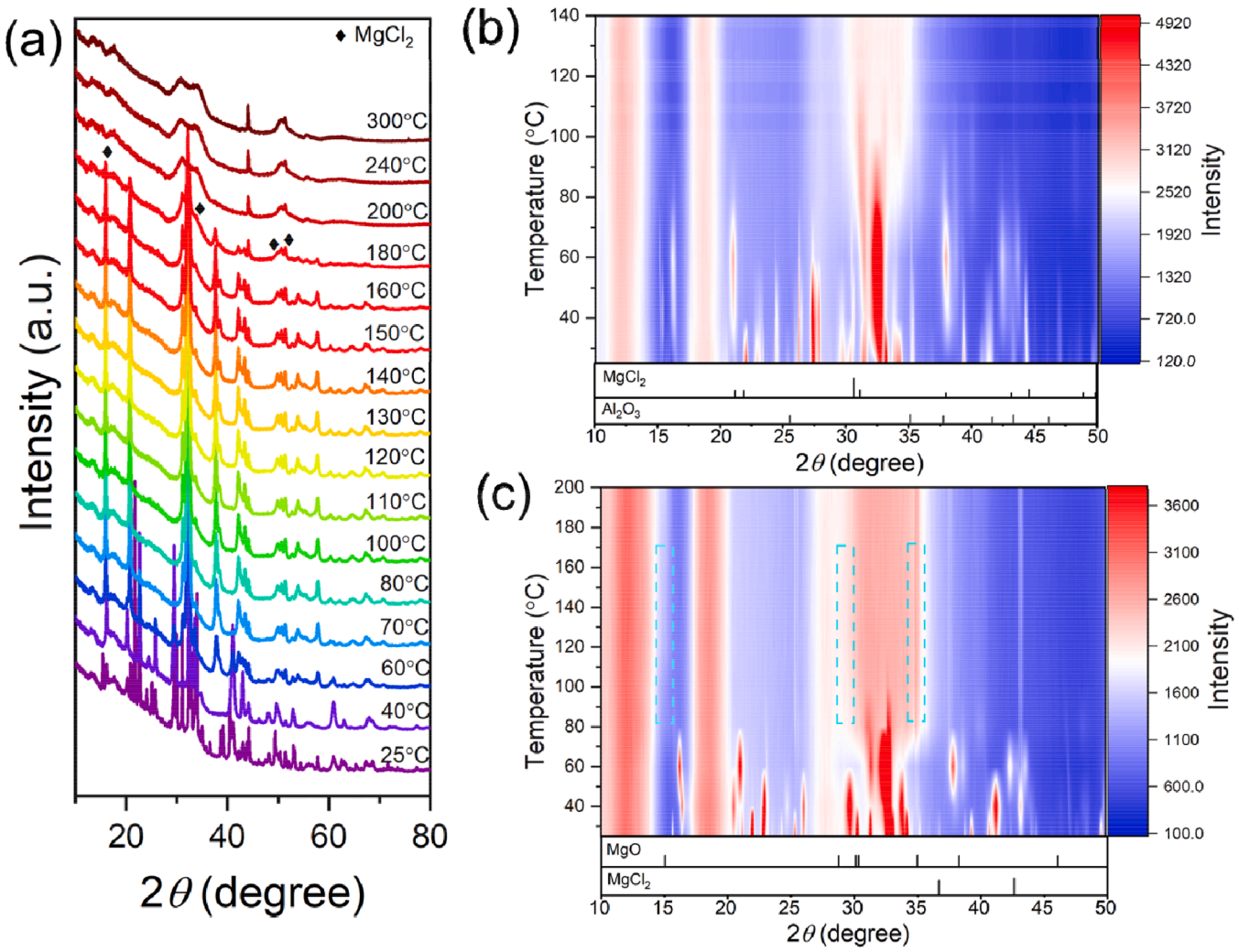


Fig. 3. In-situ high-temperature XRD patterns of (a) $MgCl_2{\cdot}6H_2O$; (b) $MgCl_2{\cdot}6H_2O/AlCl_3$; (c) $MgCl_2{\cdot}6H_2O/NH_4Cl$.

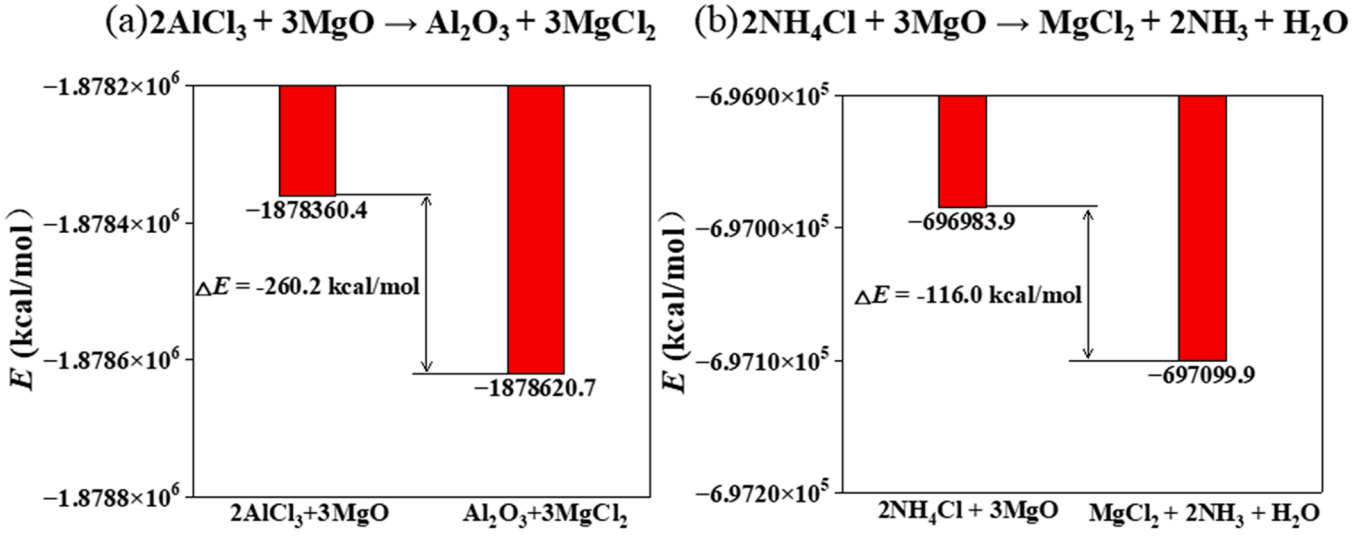


Fig. 4. Internal energy changes for the conversion reactions between MgO and $MgCl_2$.

$MgCl_2{\cdot}6H_2O/NH_4Cl$ is greater than that of $MgCl_2{\cdot}6H_2O$. The interactions between Al^{3+} or NH_4^+ and H_2O, together with the substitution of Cl^- for H_2O, result in the easier production of anhydrous $MgCl_2$. During the dehydration of $MgCl_2{\cdot}6H_2O$, several impurities including MgO are generated, indicating that the environment remains to be an important controlling factor for the dehydration of $MgCl_2{\cdot}6H_2O$. The

results obtained would be helpful for understanding the dehydration processes and mechanisms of $MgCl_2·6H_2O$ under the assistance of $AlCl_3$ and NH_4Cl, and for exploring new techniques for $MgCl_2·6H_2O$ dehydration. Follow-up work needs to focus on the separation of anhydrous $MgCl_2$ from $AlCl_3·6H_2O$, NH_4Cl, and by-products after dehydration.

CRediT authorship contribution statement

Huifang Zhang: Visualization, Validation. **Mingzhen Li:** Formal analysis. **Haining Liu:** Project administration, Methodology. **Xiushen Ye:** Supervision, Funding acquisition. **Quan Li:** Software. **Siyuan Zhang:** Investigation. **Guosheng Shi:** Resources, Project administration, Conceptualization. **Honghui Liu:** Conceptualization. **Mingzhe Dong:** Writing – review & editing. **Zhijian Wu:** Writing – review & editing, Conceptualization. **Kexin Li:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Li Han:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Data availability

Data will be made available on request.

References

[1] G.J. Kipouros, D.R. Sadoway, A thermochemical analysis of the production of anhydrous $MgCl_2$, J. Light Met. 1 (2001) 111–117, https://doi.org/10.1016/S1471-5317(01)00004-9.
[2] Z. Zhang, X. Lu, Y. Yan, A novel pathway for the preparation of Mg metal from magnesia, J. Magnes. Alloy. 10 (2022) 2847–2856, https://doi.org/10.1016/j.jma.2021.02.009.
[3] S. Bai, B. Jiang, W. He, J. Song, K. Li, Y. Yang, J. Zheng, G. Huang, D. Zhang, F. Pan, Orientation engineering of magnesium alloy: a review, J. Mater. Res. Technol. 33 (2024) 4908–4937, https://doi.org/10.1016/j.jmrt.2024.10.121.
[4] H. Shi, C. Xu, X. Hu, W. Gan, K. Wu, X. Wang, Improving the Young's modulus of Mg via alloying and compositing – a short review, J. Magnes. Alloy. 10 (2022) 2009–2024, https://doi.org/10.1016/j.jma.2022.07.011.
[5] J. Bai, Y. Yang, C. Wen, J. Chen, G. Zhou, B. Jiang, X. Peng, F. Pan, Applications of magnesium alloys for aerospace: a review, J. Magnes. Alloy. 11 (2023) 3609–3619, https://doi.org/10.1016/j.jma.2023.09.015.
[6] M. Li, C. Liu, A. Ding, C. Xiao, A review on the extraction and recovery of critical metals using molten salt electrolysis, J. Environ. Chem. Eng. 11 (2023) 109746, https://doi.org/10.1016/j.jece.2023.109746.
[7] J. Wang, J. Wang, H. Cui, Z. Li, M. Wang, W. Yi, Promotion effect of molten salt hydrate on co-esterification of biomass-derived levulinic and formic acids, Fuel 321 (2022) 124077, https://doi.org/10.1016/j.fuel.2022.124077.
[8] Q. Huang, G. Lu, J. Wang, J. Yu, Thermal decomposition mechanisms of $MgCl_2·6H_2O$ and $MgCl_2·H_2O$, J. Anal. Appl. Pyrolysis 91 (2011) 159–164, https://doi.org/10.1016/j.jaap.2011.02.005.
[9] K. Sugimoto, R.E. Dinnebier, J.C. Hanson, Structures of three dehydration products of bischofite from in situ synchrotron powder diffraction data ($MgCl_2·nH_2O$; n = 1, 2, 4), Acta Cryst. B 63 (2007) 235–242, https://doi.org/10.1107/S0108768107002558.
[10] Z. Zhang, X. Lu, F. Pan, Y. Wang, S. Yang, Preparation of anhydrous magnesium chloride from magnesium chloride hexahydrate, Metall. Mater. Trans. B 44 (2012) 354–358, https://doi.org/10.1007/s11663-012-9777-5.
[11] Z. Zhang, X. Lu, T. Wang, Y. Yan, F. Pan, The dehydration of $MgCl_2·6H_2O$ in $MgCl_2·6H_2O$-KCl-NH_4Cl system, J. Anal. Appl. Pyrolysis 110 (2014) 248–253, https://doi.org/10.1016/j.jaap.2014.09.009.
[12] R. Hamze, I. Nevoigt, U. Sazama, M. Fröba, M. Steiger, Carnallite double salt for thermochemical heat storage, J. Energy Storage 86 (2024) 111404, https://doi.org/10.1016/j.est.2024.111404.
[13] Y.-L. Wu, X.-F. Huang, M.-D. Yang, S.-P. Zou, J. Dang, H.-S. Hu, Study on the mechanisms and kinetics of complex's thermal decomposition getting anhydrous magnesium chloride, J. Anal. Appl. Pyrolysis 81 (2008) 133–135, https://doi.org/10.1016/j.jaap.2007.05.004.
[14] C. Lu, Z. Wang, Y. Zhang, G. Tang, Y. Wang, X. Guo, J. Li, L. Wei, Aqueous aluminum-zinc hybrid ion batteries with urea-based hydrated eutectic electrolytes, Nano Energy 120 (2024) 109158, https://doi.org/10.1016/j.nanoen.2023.109158.
[15] Y. Yang, N. Wang, X. Pang, A. Yasinskiy, Y. Tan, J. Yu, Z. Wang, Z. Shi, Thermodynamics of the decomposition of aluminum chloride hexahydrate to prepare alumina, J. Mater. Res. Technol. 15 (2021) 6640–6646, https://doi.org/10.1016/j.jmrt.2021.11.099.
[16] D.P. Mason, M.E. Elwood Madden, Raman spectroscopy of high salinity brines and ices, Icarus 372 (2022), https://doi.org/10.1016/j.icarus.2021.114759.
[17] W.W. Rudolph, R. Mason, C.C. Pye, Aluminium(III) hydration in aqueous solution. A Raman spectroscopic investigation and an ab initio molecular orbital study of aluminium(III) water clusters, Phys. Chem. Chem. Phys. 2 (2000) 5030–5040, https://doi.org/10.1039/b003764h.
[18] H.G.M. Edwards, D.W. Farwell, A.F. Johnson, FT-Raman spectroscopic study of aluminium(III) chloride in acetonitrile and dichloromethane solutions containing water, J. Mol. Struct. 344 (1995) 37–44, https://doi.org/10.1016/0022-2860(94)08413-C.
[19] K.S.P. Karunadasa, R.M.G. Rajapakse, H.M.T.G.A. Pitawala, C.H. Manoratne, Microstructural insight into the thermal decomposition of $MgCl_2·6H_2O$ examined by in-situ high-temperature X-ray powder diffraction, J. Solid State Chem. 322 (2023), https://doi.org/10.1016/j.jssc.2023.123965.
[20] H.-C. Eom, H. Park, H.-S. Yoon, Preparation of anhydrous magnesium chloride from ammonium magnesium chloride hexahydrate, Adv. Powder Technol. 21 (2010) 125–130, https://doi.org/10.1016/j.apt.2010.01.003.
[21] H.-H. Emons, Structure and properties of molten salt hydrates, Electrochim. Acta 33 (1988) 1243–1250, https://doi.org/10.1016/0013-4686(88)80155-5.